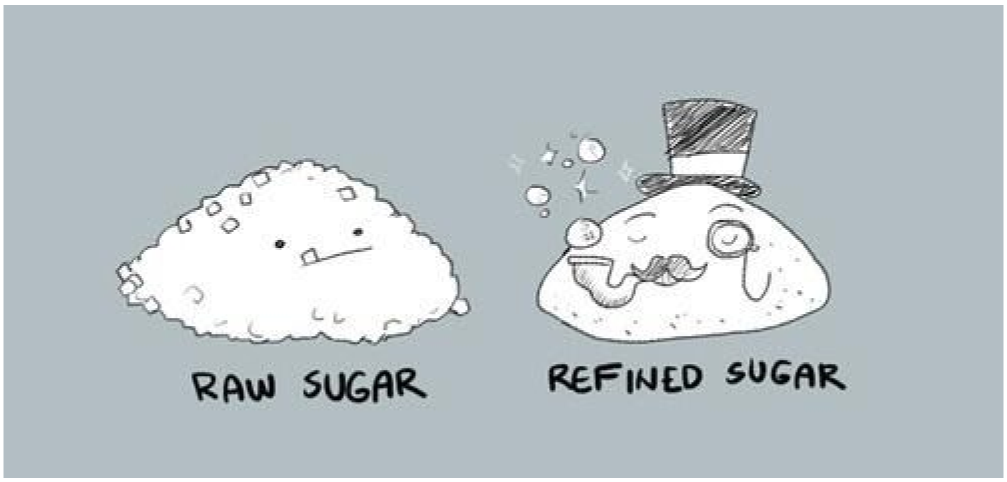
Food Science Test



Troy Invitational 2021-22

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Do NOT open test until instructed to

Make sure to:
- Write your team name and team number
- Use eye protection when it is your turn to do the lab portion
- Follow correct lab procedures and clean up properly once you have finished.
- Staple the test together before turning in if you decide to rip it apart

Questions with the asterisk will be used as tie-breakers.

Part I: Multiple Choice (60 pts)

Each question is 1 point

1. Sucrose is made up of what two monosaccharides?
    a. Glucose and galactose
    b. Glucose and ribose
    c. Glucose and fructose
    d. Glucose and glucose

2. A sugar is said to be reducing if it contains which functional group?
    a. Hydroxyl
    b. Carbonyl
    c. Carboxyl
    d. Ester

3. Which food preservation process involves heating canned foods to 100°C in order to destroy anaerobic organisms?
    a. Blanching
    b. Pasteurization
    c. Dehydration
    d. Sterilization

4. Which of the following was the first artificial sweetener to be created? (Excluding lead acetate)
    a. Mogrosides
    b. Stevia
    c. Saccharin
    d. Sucralose

5. Which of the following is a polysaccharide commonly found in cell walls of fungi?
    a. Starch
    b. Cellulose
    c. Melibiose
    d. Chitin

6. Where in the body is the enzyme pepsin located?
    a. Stomach
    b. Small intestine
    c. Colon
    d. Kidney

7. Which of the following is an enzyme that breaks down proteins into amino acids and is usually found in the small intestines?
    a. Kinase
    b. Chymosin
    c. Pepsin
    d. Trypsin

8. Oxidoreductases are enzymes that catalyze the transfer of
    a. Hydrogen ions
    b. Protons
    c. Electrons
    d. Neutrons

9. Which of the following is a pentose sugar?
    a. Deoxyribose
    b. Fructose
    c. Maltose
    d. Mannose

10. Which of the following is NOT an example of a sugar alcohol?
    a. Sorbitol
    b. Butanol
    c. Xylitol
    d. Lactitol

11. Which of the following organisms is able to survive foods with a water activity value of 0.93?
    a. E. coli
    b. C. perfringens
    c. C. botulinum
    d. L. monocytogenes

12. Which of the following is a food with an a_w value of 0.99?
    a. Milk
    b. Salami
    c. Raw meats
    d. Distilled water

13. What are the products of glycolysis per glucose?
    a. 2 ATP, 2 NADH, 2 pyruvate
    b. 3 ATP, 2 NADH, 4 pyruvate
    c. 1 ATP, 4 $FADH_2$, 2 pyruvate
    d. 4 ATP, 2 NADH, 1 pyruvate

14. Another name of the citric acid cycle is
    a. Crabs cycle
    b. Tricarboxylic acid cycle
    c. Acetyl CoA cycle
    d. None of the above

15. What glycosidic linkages does cellulose contain?
    a. Alpha 1-4
    b. Beta 1-4
    c. Gamma 1-4
    d. Delta 1-4

16. Why was the artificial sweetener sucralose banned from the US?
    a. Lead to higher rates of heart disease
    b. Was shown to break down into carcinogens once inside the body
    c. Was shown to cause liver cancer in mice
    d. Sucralose is not banned in the US

17. Which food preservation process involves storing foodstuff at temperatures of 4°C and below?
    a. Freezing
    b. Confit
    c. Refrigeration
    d. Dehydration

18. Which of the following is/are benefits of preserving foods through burials?
    a. Lack of oxygen
    b. Cool temperatures
    c. Lack of light
    d. All of the above

19. The citric acid cycle is the only process within cellular respiration that produces what as a byproduct?
    a. H_2O
    b. CO_2
    c. NADH
    d. ATP

20. The Maillard reaction is responsible for the browned color in cooked foods such as steaks and bread. A chemical reaction between what two compounds causes the Maillard reaction to occur?
    a. Proteins and nucleic acids
    b. Saturated fats and carbohydrates
    c. Cholesterols and enzymes
    d. Amino acids and reducing sugars

21. Which of the following is true for Intermediate Moisture Foods? Select all that applies.
    a. They are shelf-stable products
    b. Have a moisture content ranging from 45% - 60%
    c. They are not susceptible to yeast and mold growth.
    d. Examples include dried fruits, marshmallows, and chips
    e. Have water activities of 0.6-0.85

22. Generally, water activity dictates that moisture will migrate from a region of ____ aw to a region of ____ aw.
    a. High, lower
    b. Low, higher
    c. All of the above
    d. Neither

23. Match the following letters with the appropriate choice. The letters could be used more than once.

A. 0.9 B. 0.8 C. 0.75 D. 0.65 E. 0.60

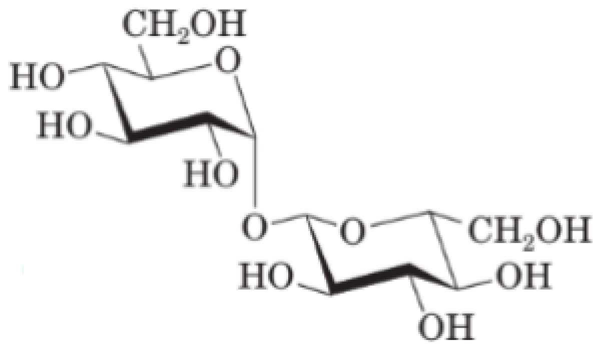
1. Bacteria stops growing at ______ aw
2. Yeast stops growing at ______ aw
3. Molds stops growing at ______ aw
4. Halophilic Bacteria ______ aw
5. Xerophilic Moulds ______ aw
6. Osmophilic Moulds ______ aw
7. Lowest limit is ______ aw

24. How is water activity measured? Select all that applies.
    a. Resistive Electrolytic Hygrometers (REH)
    b. Capacitance Hygrometers
    c. Dew Point Hygrometers
    d. Chilled mirror
    e. No existing device can measure water activity

25. What is the phenomenon by which Saccharomyces cerevisiae produces ethanol in aerobic conditions rather than producing biomass via the TCA cycle? Select all that applies.
    a. Crabtree Effect
    b. Catabolite repression
    c. Fermentation
    d. Warburg effect
    e. Pasteur effect
    f. None of the above

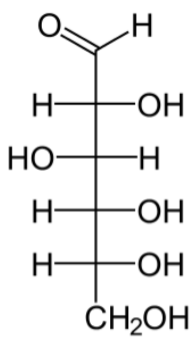
26. What is the phenomenon by which tumor cells favor glycolysis over the oxidative phosphorylation pathway? Select all that applies.
    a. Crabtree Effect
    b. Catabolite repression
    c. Fermentation
    d. Warburg effect
    e. Pasteur effect
    f. None of the above

27. Identify the following molecule.



a. Sucrose
b. Lactose
c. Trehalose
d. Maltose
e. Cellobiose

28. Identify the following molecule.



a. D-glucose
b. L-glucose
c. D-galactose
d. L-galactose
e. D-fructose
f. L-fructose

29. Enzymatic browning is caused by which enzyme? Select all that applies.

a. Phenylalanine ammonia-lyase
b. Superoxide dismutase
c. Alkaline phosphatase
d. Polyphenol oxidase
e. None of the above

30. Competitive inhibitors have their Vmax _______ & Km _______.
    a. Increased, unaffected
    b. Decreased, unaffected
    c. Unaffected, decreased
    d. Increased, decreased
    e. Decreased, increased
    f. Unaffected, increased

31. Noncompetitive inhibitors have their Vmax _______ & Km _______.
    a. Increased, unaffected
    b. Decreased, unaffected
    c. Unaffected, decreased
    d. Increased, decreased
    e. Decreased, increased

32. What is the SI base unit of density?
    a. kg/m^3
    b. g/m^3
    c. $L^{-3}M$
    d. lb/ft^3
    e. None of the above

33. What are the 6 types of enzymes? Select all that applies.
    a. Hydrolases
    b. Oxidoreductases
    c. Helicases
    d. Lyases
    e. Polymerases
    f. Transferases
    g. Ligases
    h. Isomerases
    i. Catalases
    j. Coenzymes

34. *What three reactions in gluconeogenesis are so energetically unfavorable that the reactions favoring glycolysis are effectively irreversible? **TB #2**
    a. Conversion of pyruvate to PEP.
    b. Conversion of fructose-1,6-bP to fructose-6-P
    c. Conversion of DHAP to fructose 1,6-bP
    d. Conversion of glucose-6-P to glucose
    e. Conversion of fructose-6-P to glucose-6-P

35. If a sphere is heated, the volume of the sphere will _______ & the density will _________.
    a. Decrease, increase
    b. Stay unaffected, decrease
    c. Stay unaffected, stay unaffected
    d. Increase, decrease
    e. None of the above

36. Although much of the protein in meats is lost through cooking, it is important to cook eggs because… (select all that apply)
    a. It helps with digestibility
    b. It gains protein when cooked, which is why many people eat them to become strong
    c. It isn't as sensitive to protein loss
    d. It kills a lot of the bacteria
    e. None of the above

37. What is the main purpose of glycolysis and citric acid cycle in cellular respiration?
    a. To make ATP
    b. To use glucose
    c. To make NADH
    d. A and C
    e. All of the above
    f. None of the above

38. Which of the following is not part of the 8 main allergens?
    a. Soybeans
    b. Peanuts
    c. Fish
    d. Gluten
    e. All of the above are the most common allergens.

Match the following macromolecules to their respective tests. Write the letter on the blank. (Some blanks may or may not have more than one letter.)
a. Biuret test b. Iodine test c. Ethanol emulsion test d. Sudan IV e. Benedict's test

39. Reducing sugars ________
40. Starches ________
41. Proteins ________
42. Lipids ________

Match the following food tests with their positive reactions. Not all letters will be used.
a. red/orange b. Dark green c. purple/lilac d. cloudy white e. Black f. Dark red g. blue

43. Biuret test ________
44. Iodine test ________
45. Ethanol emulsion test ________
46. Sudan IV ________
47. Benedict's test ________

48. True or false: The main difference between essential and non-essential amino acids is that essential is important for functions in the body, while non-essential amino acids aren't.

49. Which of the following is not an essential amino acid?
    a. Tryptophan
    b. Tyrosine
    c. Threonine
    d. None of the above

50. Which of the following is not part of the 3 main omega-3 fatty acids?
    a. alpha-linolenic acid (ALA)
    b. eicosapentaenoic acid (EPA)
    c. docosahexaenoic acid (DHA)
    d. n-hexadecanoic acid (NDA)
    e. More than one answer is correct

51. Bob lost control of his body movements and feeling in his arms and legs. He also has a weakened immune system. Which vitamin is Bob lacking?
    a. Vitamin A
    b. Vitamin B
    c. Vitamin C
    d. Vitamin D
    e. Vitamin E
    f. None of the above

52. Which sugar is sweetest in the options below?
    a. Sucrose
    b. Honey
    c. Aspartame
    d. Sucralose

53. What plant(s) are molasses made from? Select all that apply.
    a. Sugar beets
    b. Sugar canes
    c. Sorghum plant
    d. Metopium brownei

54. Which of the following does cream of tartar do?
    a. Stabilize whipped egg whites
    b. Prevent sugar from crystallizing
    c. Act as a leavening agent for baked goods
    d. Only A and C
    e. All of the above
    f. None of the above

55. FDA recent Sodium Reduction Final Guidance aims to lower sodium intake to _____ per day.
    a. 2500 mg
    b. 3000 mg
    c. 3200 mg
    d. 1500 mg
    e. 1800 mg

56. True or false: calcium and Vitamin D are required on both the old and new nutrition labels.

57. What is mother liquor?
    a. An alcohol that has been fermented by yeast and aged for many years.
    b. A byproduct of irradiation of certain vegetables.
    c. A solution remaining after a component has been removed by some process.
    d. None of the above

58. True or false: A solution must always be supersaturated to form crystals.

59. Which of the following are factors that affect the rates of nucleation and crystal growth?
    a. Temperature
    b. Degree of supersaturation
    c. Interfacial tension between the solute and the solvent
    d. B and C only
    e. All of the above

60. * Bob wants to make nougats. He cooks his sugar syrup and measures the temperature, then concludes that it is 87% sugar concentration. How much hotter does he have to cook it to make nougats? **TB #4**
    a. 5-20℉
    b. 10-30℉
    c. 15-40℉
    d. 25-50℉
    e. He needs to make a new batch to lower the temperature.

Part II: Written Portion (20 pts)

Each question is 1 point.

Use the nutrition label below to answer questions 1-7.

Nutrition Facts

Serving size 1 potato (148g/5.2oz)

Amount per serving

Calories

	% Daily Value*
Total Fat 0g	**0%**
Saturated Fat 0g	**0%**
Trans Fat 0g	
Cholesterol 0mg	**0%**
Sodium 0mg	**0%**
Total Carbohydrate	**9%**
Dietary Fiber 2g	**7%**
Total Sugars 1g	
Includes 0g Added Sugars	**0%**
Protein 3g	
Vitamin D 0mcg	0%
Calcium 20mg	2%
Iron	6%
Potassium 620mg	15%
Vitamin C 27mg	30%
Vitamin B_6 0.2mg	10%

* The % Daily Value (DV) tells you how much a nutrient in a sering of food contributes to a daily diet. 2,000 calories a day is used for general nutrition advice.

1. If one serving of this product provides 104 kcal of energy solely from carbohydrates, how many grams of total carbohydrates does one serving of this product contain? (Make sure to include units!)

   ____

2. If the recommended daily value for iron is 18.3 mg, how much iron does one serving of this product contain? (Round to the nearest tenth and make sure to write units!)

   ____

3. How many *calories* does one serving of this product contain? (Make sure to include units!)

   ____

4. Based on the nutrition label above, what would the daily value for potassium in mg be? (Round to the nearest integer and make sure to include units!)

   ____

5. For one serving of this product, how many Calories come from protein? (Make sure to include units!)

   ____

6. * How many kilojoules (kJ) of energy do 3 servings of this product contain? (Round to the nearest integer and include units!) **TB3**

   ____

7. List all vitamins that are lipophilic.

   ____

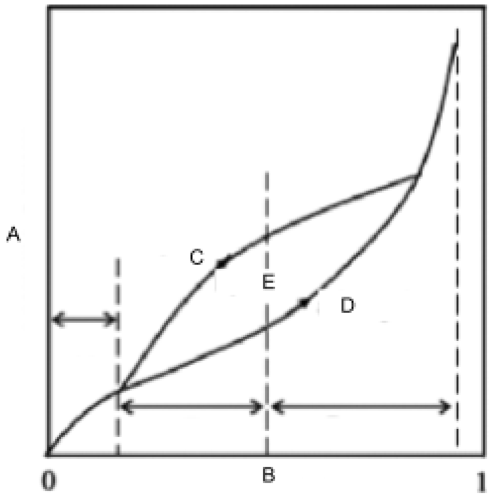
8. What is the difference between moisture content and water activity?

   ____

   ____

9. What is this diagram called? Fill in the following diagram relating moisture content and water activity.

   ____



A. ___________________________
B. ___________________________
C. ___________________________
D. ___________________________
E. ___________________________

10. What are the similarities and differences between osmophiles, xerophiles, and halophiles?

11. Write the generalized equation for enzymatic reactions.

____

12. What are zymogens?

____

13. Name and describe the model that describes enzyme-substrate interaction.

____

____

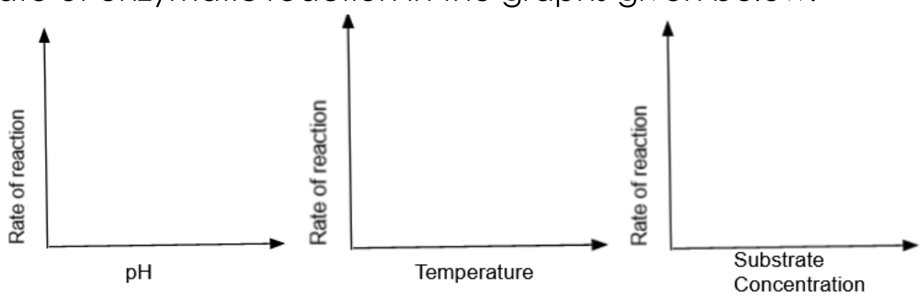
14. If you have a graduated cylinder filled with water and a scale, how can you determine the density of an irregular shaped object like the one shown below?

____

____

____

15. Draw the effect of pH, temperature, and substrate concentration on the rate of enzymatic reaction in the graphs given below.



16. Determine the density in g/L of an unknown with the following information. A 55 gallon tub weighs 137.5lb when empty and 500.0 lb when filled with the unknown.

____

17. How many kilograms of mercury would fill a 5-liter container if the density of mercury is 13.6 grams/centimeter^3?

____

18. What is the mass of a cylinder of lead that is 2.50 cm in diameter, and 5.50 cm long. The density of lead is 11.4 g/mL and the volume of a cylinder is .

____

19. Bob was making soup. He put in carrots, potatoes, beef, heavy cream, olive oil, and water. After everything is boiled, he notices that there is foam at the top of his soup. Which of the ingredients caused this and explain why.

20. Bob wanted to make light and "fluffy" cookies. However, he did not have any baking powder, so he used only baking soda as a substitute. As a result, he finds that his cookies are thick and dense instead. Explain why.

Part II: Lab (20 pts)

You will be given __ minutes to finish all 3 labs. Inability to clean up after time runs out will result in a penalty.

1. * Using a hydrometer, determine the sugar content of the given sugar solution. Write your answer below. **TB1** (5 pts)

   ____

2. Given sugar solutions A and B, use the given Benedict's solution and hot water bath to determine whether each solution has a reducing sugar or not. In the spaces below, write "Yes" or "Reducing" if the solution contains a reducing sugar and write "No" or "Not reducing" if the solution does not contain a reducing sugar. (10 pts, 5 pts for each solution)

   Solution A: ____

   Solution B: ____